Image Analysis and Computer Vision

Homework Report

Pietro Filippo Schgor
Person code: 10710527

January 18, 2026

1 Introduction

This report presents a geometric analysis and 3D reconstruction based on a single image of the San Maurizio al Monastero Maggiore church. Starting from a set of manually extracted features, we perform a stratified rectification of the vault, calibrate the camera, and reconstruct the 3D geometry of the architectural arches and the cylinder axis.

2 Feature Extraction and Vanishing Geometry (Q1 & Q2)

Feature Extraction Strategy Feature extraction was performed manually to prioritize accuracy, given the challenges in detecting elements like arcs amidst the high levels of noise introduced by the scene's decorations. Furthermore, all extracted features have been stored in dedicated files to ensure the reproducibility of the results. The image was first resized by a factor of 0.7 to reduce computational load.

(a) (b)

Figure 1: Given and extracted segments

Q1: Vanishing Line of the Horizontal Plane

Theory: The vanishing line (horizon) is the image of the line at infinity ℓ_∞, representing the set of points at infinity. To identify the vanishing line of the plane perpendicular to the cylinder axis, we compute the vanishing points derived from sets of parallel lines in both the X and Y directions 1:

- $\mathbf{V}_x$ (Vanishing Point X): The intersection of the set of lines $\mathcal{L} = \{l_1, l_2, l_3\}$ parallel to the X-axis in 3D space.

- $\mathbf{V}_y$ (Vanishing Point Y): The intersection of the set of lines $\mathcal{M} = \{m_1, m_2, m_3\}$ parallel to the Y-axis in 3D space.

The line connecting $\mathbf{V}_x$ and $\mathbf{V}_y$ defines the vanishing line of the horizontal plane, denoted as ℓ'_∞:

$$\ell'_\infty = \mathbf{V}_x \times \mathbf{V}_y = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \tag{1}$$

Results: The set of lines $\mathcal{L} = \{l_1, l_2, l_3\}$, $\mathcal{M} = \{m_1, m_2, m_3\}$ have been extracted selecting a pair of points for every line and computing the cross product. The vanishing points $\mathbf{V}_x$, $\mathbf{V}_y$ have been computed; the resulting vanishing line is:

$$\ell'_\infty = \mathbf{V}_x \times \mathbf{V}_y = \begin{bmatrix} -0.0011 \\ -0.0003 \\ 1.0000 \end{bmatrix} \tag{2}$$

Q2: Vanishing Point of the Cylinder Axis

Theory: We compute the vanishing point in the direction of the cylinder axis as the intersection of a pair of lines parallel to the aforementioned axis. Let $\mathbf{V}_z$ be the Vanishing Point Z: it is the intersection of the set of lines $\mathcal{W} = \{w_1, w_2, w_3\}$ parallel to the Z-axis in 3D space.

Results: The set of lines $\mathcal{W} = \{w_1, w_2, w_3\}$ have been extracted. The vanishing point $\mathbf{V}_z$ is computed as the cross product of the pair of more distant lines in $\mathcal{W}$ to minimize the risk of selecting lines apparently parallel.

$$\mathbf{V}_z = \begin{bmatrix} 6198.5 \\ 2474.4 \\ 1.000 \end{bmatrix} \tag{3}$$

3 Rectification (Q3)

Methodology Using both the results of the previous points and the image of some of the diagonal arcs, we find a Euclidean rectification mapping $\boldsymbol{H}_R$ for one of the vertical planes perpendicular to the cylinder axis. I chose the *Stratified Rectification* method starting from the circular constraint, as there are not at least two non-parallel independent pairs of orthogonal lines available.

3.1 Affine Reconstruction

Theory: We construct the affine mapping $\boldsymbol{H}_A$ that maps the imaged vanishing line ℓ'_∞ to the canonical line at infinity $\mathbf{l}_\infty = (0, 0, 1)^\top$:

$$\boldsymbol{H}_A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ a & b & c \end{bmatrix}, \boldsymbol{H}_A^{-T} \ell'_\infty = \mathbf{l}_\infty \tag{4}$$

This transformation produces an image where the sets $\mathcal{L}$ and $\mathcal{M}$ become parallel. However, the image remains affinely distorted (angles are not preserved) 2.

Results: We compute the affine mapping $\boldsymbol{H}_A$ that maps the imaged vanishing line ℓ'_∞ to the canonical line at infinity:

$$\boldsymbol{H}_A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -0.0011 & -0.0003 & 1.0000 \end{bmatrix}$$

Figure 2: Affine Rectification

3.2 Metric Reconstruction (From Affine to Metric)

Theory: We leverage the knowledge of the vault's geometry: the intersection of the cylindrical vault with a plane $X-Y$ perpendicular to the axis is a semicircle. We determine a second affine transformation $\boldsymbol{H}_B$ to rescale the rectified image, ensuring the axes of the circle are perpendicular and of equal length.

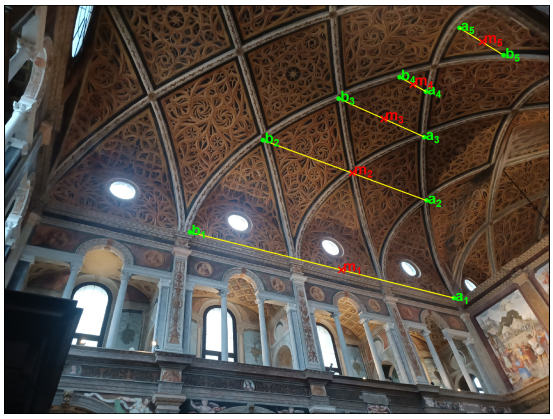


Figure 3: Midpoints

a) Feature Extraction (Harmonic Conjugates): Due to the symmetry of the diagonal arches, the midpoint of each pair of symmetric endpoints (a_i, b_i) 3 lies on a plane perpendicular to the cylinder axis. In the image, the midpoint m_i is obtained as the harmonic conjugate of the vanishing point $\mathbf{V}_z$ with respect to the points a_i and b_i, satisfying the cross-ratio:

$$(a_i, b_i; m_i, V_z) = -1$$

This construction is invariant under projective transformations and allows the recovery of the midpoints up to scale.

b) Conic Fitting: We fit a conic section to these midpoints using the Direct Linear Transformation (DLT) method. The coefficient vector $\mathbf{c}$ minimizes $\|\boldsymbol{D}\mathbf{c}\|$ (via SVD) for the conic equation:

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0 \tag{5}$$

Results: To compute $\boldsymbol{H}_B$ we extracted a large set of midpoints (N=12) to ensure a stable

conic fitting. 4. The conic $\mathbf{C}$ fitted on these points is:

$$\mathbf{C} = \begin{bmatrix} 1.2647 \times 10^{-7} & 4.3709 \times 10^{-8} & -3.5206 \times 10^{-4} \\ 4.3709 \times 10^{-8} & 1.0981 \times 10^{-7} & -2.0902 \times 10^{-4} \\ -3.5206 \times 10^{-4} & -2.0902 \times 10^{-4} & 1.0000 \end{bmatrix}$$

Figure 4: Conic Fitting

c) Metric Transformation Construction: Let $\boldsymbol{Q}$ be the matrix of the affine-rectified conic:

$$\boldsymbol{Q} = \boldsymbol{H_A}^{-T} \boldsymbol{C} \boldsymbol{H_A}^{-1} \tag{6}$$

We extract the geometric properties of the ellipse from $\boldsymbol{Q}$: orientation α, semi-major axis a, and semi-minor axis b. To restore circularity, we decompose the transformation $\boldsymbol{G}$ into rotation and isotropic scaling:

Let $\boldsymbol{R}_\alpha$ be the rotation matrix and $\boldsymbol{S}$ the scaling matrix (with $\sigma = a/b$):

$$\boldsymbol{R}_\alpha = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix}, \quad \boldsymbol{S} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{a}{b} \end{pmatrix} \tag{7}$$

The local affine deformation $\boldsymbol{G}$ is:

$$\boldsymbol{G} = \boldsymbol{R}_\alpha \cdot \boldsymbol{S} \cdot \boldsymbol{R}_\alpha^T \tag{8}$$

Embedding this into a homography $\boldsymbol{H}_B$:

$$\boldsymbol{H}_B = \begin{bmatrix} G_{11} & G_{12} & 0 \\ G_{21} & G_{22} & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{9}$$

The final Metric Rectification Homography $\boldsymbol{H}_R$ is:

$$\boldsymbol{H}_R = \boldsymbol{H}_B \cdot \boldsymbol{H}_A \tag{10}$$

Now we can compute the position on the rectified plane $\mathbf{x_R}$ of a non-apical nodal point $\mathbf{x}$ as follows:

$$\mathbf{x_R} = \boldsymbol{H_R} \cdot \mathbf{x} \tag{11}$$

Results: From this result, we can compute $\boldsymbol{H}_B$ circularizing the conic [3]:

$$\mathbf{H_B} = \begin{bmatrix} 0.9916 & 0.0760 & 0 \\ 0.0760 & 0.3118 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The final metric rectification is:

$$H_R = H_B \cdot H_A = \begin{bmatrix} -0.9916 & -0.0760 & 0 \\ -0.0760 & -0.3118 & 0 \\ -0.0012 & -0.0003 & 1 \end{bmatrix}$$

We can see that in the metric rectified image the angles are preserved 5.

Figure 5: Metric Rectification

We take a non-apical nodal point $\mathbf{m_2}$ 3 from the midpoints previously extracted:

$$\mathbf{m_2} = \begin{bmatrix} 1800.9 \\ 845.6 \\ 1.0 \end{bmatrix}$$

Therefore the position on the rectified plane:

$$\mathbf{m_{2,R}} = \begin{bmatrix} -1336.2 \\ -327.8 \\ 1.0 \end{bmatrix}$$

4 Camera Calibration (Q4)

Theory From the results of the previous points, we compute the calibration matrix $\mathbf{K}$ of the camera. The internal calibration of the camera is encapsulated in the matrix $\boldsymbol{K}$. To retrieve it, we estimate the *Image of the Absolute Conic* (IAC), denoted as $\boldsymbol{\omega}$. The relationship is given by:

$$\boldsymbol{\omega} = (\boldsymbol{K}\boldsymbol{K}^T)^{-1} \tag{12}$$

We proceed with the assumption of **zero skew** ($s = 0$), which implies that the camera sensor axes are orthogonal. The matrix $\boldsymbol{\omega}$ takes the following form:

$$\boldsymbol{\omega} = \begin{bmatrix} \omega_{11} & 0 & \omega_{13} \\ 0 & \omega_{22} & \omega_{23} \\ \omega_{13} & \omega_{23} & \omega_{33} \end{bmatrix} \tag{13}$$

To solve for the unknown entries of $\boldsymbol{\omega}$, we construct a linear system $\boldsymbol{A}\mathbf{w} = \mathbf{0}$ based on the geometric constraints of the scene:

1. **Constraints from Rectifying Homography (Zhang's Constraints):**
   Let $\boldsymbol{H} = [\mathbf{h}_1, \mathbf{h}_2, \mathbf{h}_3]$ be the inverse of the rectification homography ($\boldsymbol{H} = \boldsymbol{H}_R^{-1}$). Since the rectification maps the image to a metric plane (where basis vectors are orthogonal and of equal length), the columns $\mathbf{h}_1$ and $\mathbf{h}_2$ must satisfy:

$$\mathbf{h}_1^T \boldsymbol{\omega} \mathbf{h}_2 = 0 \quad \text{(Orthogonality)} \tag{14}$$

$$\mathbf{h}_1^T \boldsymbol{\omega} \mathbf{h}_1 - \mathbf{h}_2^T \boldsymbol{\omega} \mathbf{h}_2 = 0 \quad \text{(Equal Magnitude)} \tag{15}$$

2. **Constraints from Vanishing Points (Pole-Polar relationship):**
   The vertical vanishing point $\mathbf{V}_z$ corresponds to the direction normal to the ground plane, corresponding to the *pole* of the vanishing line of the plane (ℓ'_∞) with respect to the IAC:

$$\ell'_\infty \propto \boldsymbol{\omega} \mathbf{V}_z \tag{16}$$

   This provides additional linear equations (via cross-product $(\boldsymbol{\omega}\mathbf{V}_z) \times \ell'_\infty = \mathbf{0}$) to constrain the solution.

Results The system is solved via Singular Value Decomposition (SVD) to find the parameter vector $\mathbf{w}$. The image of the absolute conic ω is:

$$\omega = \begin{bmatrix} 1.2540 \times 10^{-7} & 0 & -1.8899 \times 10^{-4} \\ 0 & 1.2540 \times 10^{-7} & -1.3506 \times 10^{-4} \\ -1.8899 \times 10^{-4} & -1.3506 \times 10^{-4} & 1.0000 \end{bmatrix}$$

As we expected, ω is symmetric and positive definite, therefore we can apply the Cholesky decomposition to its inverse to compute $\mathbf{K}$:

$$K = \begin{bmatrix} 2131.4695 & 0 & 1507.0468 \\ 0 & 2131.4695 & 1076.9679 \\ 0 & 0 & 1 \end{bmatrix}$$

We observe that $f_x = f_y = 2131.4695$ therefore the pixel ratio (1:1) is preserved. We also observe that the Principal point ($u_0 = 1507.0468, v_0 = 1076.9679$) is really close to the center of the image 6.

Figure 6

5 3D Reconstruction (Q5)

Methodology Using the results of the previous points, we choose one of the diagonal arcs and compute the 3D coordinates of a dozen of its points. The 3D reconstruction is performed

Figure 7

by exploiting the geometric relationship between two parallel diagonal arcs, denoted as arc $\mathcal{A}$ and arc $\mathcal{B}$, the vanishing point of the cylinder axis $\mathbf{v}_z$, and the known Euclidean distance $d = 1$ between the two parallel arcs, we can triangulate the depth of corresponding points using the Law of Sines.

The procedure is as follows:

1. **Back-projection:** Let $\mathbf{u}_A$ and $\mathbf{u}_B$ be the projections of corresponding points on the two arcs. We back-project them into unit viewing rays in the camera reference frame:

$$\mathbf{d}_A = \frac{\boldsymbol{K}^{-1}\mathbf{u}_A}{\|\boldsymbol{K}^{-1}\mathbf{u}_A\|}, \quad \mathbf{d}_B = \frac{\boldsymbol{K}^{-1}\mathbf{u}_B}{\|\boldsymbol{K}^{-1}\mathbf{u}_B\|} \tag{17}$$

2. **Axis Direction:** The 3D direction of the cylinder axis (and thus the direction connecting the two parallel arcs) is given by the back-projection of the vanishing point $\mathbf{v}_z$:

$$\mathbf{n} = \frac{\boldsymbol{K}^{-1}\mathbf{v}_z}{\|\boldsymbol{K}^{-1}\mathbf{v}_z\|} \tag{18}$$

3. **Triangulation (Law of Sines):** We solve the triangle formed by the camera center O and the 3D points A and B. We define the angles:

$$\theta = \arccos(\mathbf{d}_A \cdot \mathbf{d}_B), \quad \alpha = \arccos(\mathbf{d}_A \cdot \mathbf{n}) \tag{19}$$

The third angle is $\beta = \pi - (\theta + \alpha)$. Using the Law of Sines with the known baseline d:

$$\|A\| = d \cdot \frac{\sin(\beta)}{\sin(\theta)}, \quad \|B\| = d \cdot \frac{\sin(\alpha)}{\sin(\theta)} \tag{20}$$

4. **3D Coordinates:** The final coordinates are obtained by scaling the viewing rays:

$$\mathbf{A}_{3D} = \|A\| \cdot \mathbf{d}_A \tag{21}$$

Results The 3D reconstruction of one of the diagonal arches was performed by leveraging the known distance $d = 1$ between parallel diagonal arches. Twelve pairs of parallel points on respective parallel arches were extracted 7. We can see the reconstruction of the 3D points of the arc composed by the set of points $\{a_i\}$ in the images 9 in the last page . The 3D coordinates of the points are reported in the following table 1.

Table 1: Reconstructed 3D coordinates of the selected diagonal arc.

Point	X	Y	Z
1	0.0034	0.3357	3.6998
2	0.0499	0.1367	3.8859
3	0.1301	-0.0126	3.9814
4	0.2582	-0.1645	4.0926
5	0.3954	-0.2958	4.1128
6	0.5686	-0.4506	4.1867
7	0.7928	-0.5800	4.2011
8	1.0165	-0.6938	4.1842
9	1.2554	-0.7891	4.1327
10	1.5479	-0.8806	4.0263
11	1.8077	-0.9435	3.9770
12	2.0721	-0.9676	3.8430

Figure 8: selected diagonal arch

6 Cylinder Axis Localization (Q6)

Theory Using $\boldsymbol{K}$, we localize the cylinder axis w.r.t. the camera. The cylinder axis is a 3D line parallel to the direction $\mathbf{n}$ (computed in Q5) passing through the geometric center of the vault's cross-section. To localize it, we combine the pole-polar relationship in the image plane with 3D plane intersection.

1. **Image Center Localization:** The center of the circular cross-section in the image plane, denoted as $\mathbf{x}_c$, corresponds to the *pole* of the vanishing line ℓ'_∞ with respect to the fitted conic $\boldsymbol{C}$ (from Q3).
$$\mathbf{x}_c = \boldsymbol{C}^{-1}\ell'_\infty \tag{22}$$
This point represents the projection of the axis centers at infinity.

2. **Back-projection:** We generate the viewing ray $\mathbf{r}_c$ passing through the camera center and the image center $\mathbf{x}_c$:
$$\mathbf{r}_c = \boldsymbol{K}^{-1}\mathbf{x}_c \tag{23}$$
The physical 3D axis must intersect this ray.

3. **3D Anchor Point:** To resolve the scale ambiguity (i.e., where along the ray $\mathbf{r}_c$ the axis is located), we define a reference plane π_{ref} passing through one of the reconstructed points from Q5 (e.g., the point $\mathbf{a_5}$) and perpendicular to the axis direction $\mathbf{n}$. The plane equation is $\mathbf{n}^T(\mathbf{X} - \mathbf{a_5}) = 0$.

4. **Intersection:** The 3D center point $\mathbf{C}_{3D}$ is found by intersecting the ray $\mathbf{r}_c$ with the plane π_{ref}. Substituting $\mathbf{X} = \lambda \mathbf{r}_c$ into the plane equation:

$$\lambda = \frac{\mathbf{n}^T \mathbf{a_5}}{\mathbf{n}^T \mathbf{r}_c} \implies \mathbf{C}_{3D} = \lambda \mathbf{r}_c \tag{24}$$

5. **Axis Equation:** The cylinder axis is finally described as the line passing through $\mathbf{C}_{3D}$ with direction $\mathbf{n}$:

$$\mathcal{L}_{axis}(t) = \mathbf{C}_{3D} + t \cdot \mathbf{n} \tag{25}$$

Results We selected a random point of the arc of the previous point, so we already have its 3d coordinates:

$$\mathbf{a_5} = \begin{bmatrix} 0.3954 \\ -0.2958 \\ 4.1128 \end{bmatrix}$$

we compute the coordinates for the center of the circular cross-section:

$$\mathbf{C_{3D}} = \begin{bmatrix} 1.1086 \\ -0.1790 \\ 2.4665 \end{bmatrix}$$

At last the cylinder axis:

$$\mathcal{L}_{axis}(t) = \begin{bmatrix} 1.1086 \\ -0.1790 \\ 2.4665 \end{bmatrix} + t \cdot \begin{bmatrix} 0.8787 \\ 0.2617 \\ 0.3992 \end{bmatrix}$$

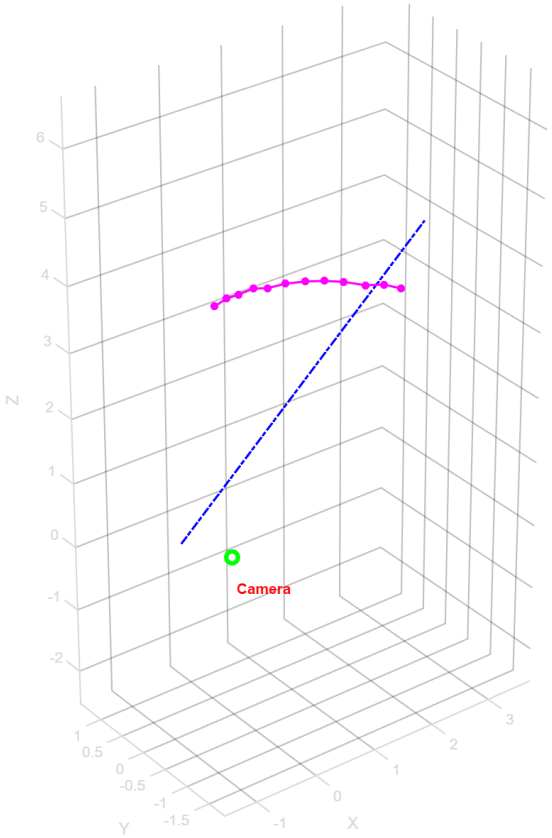
References

[1] Vincenzo Caglioti. *2D Projective Geometry and 2D Reconstruction.* Image Analysis and Computer Vision - Course Slides. Politecnico di Milano, 2025.

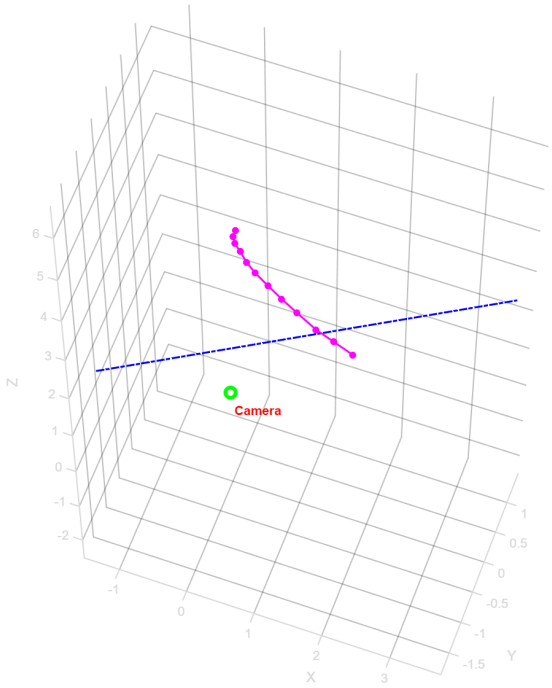
[2] Vincenzo Caglioti. *Single-view Geometry and Camera Calibration.* Image Analysis and Computer Vision - Course Slides. Politecnico di Milano, 2025.

[3] Luca Magri. *Image Analysis and Computer Vision - Metric rectification stratified.* `https://magrilu.github.io/iacv/`. Politecnico di Milano. 2025.

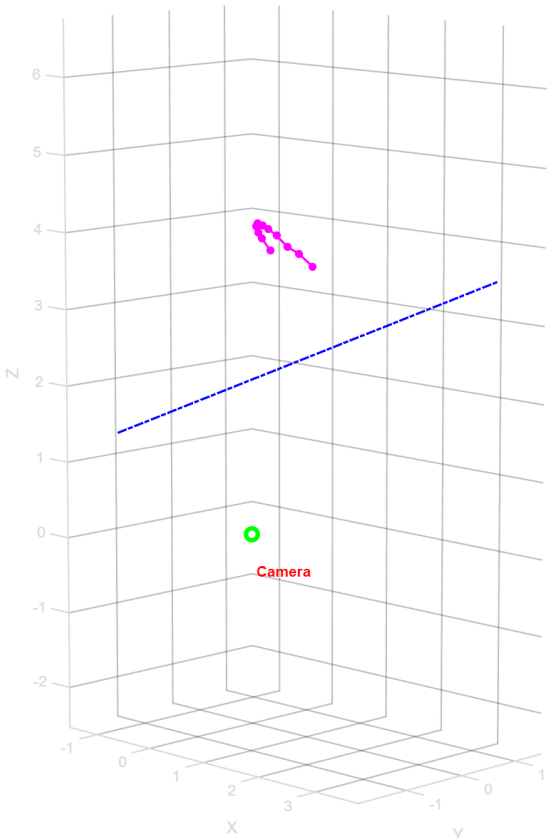
Figure 9: Multi-view visualization of the 3D reconstructed arc, the camera position, and the cylinder axis.



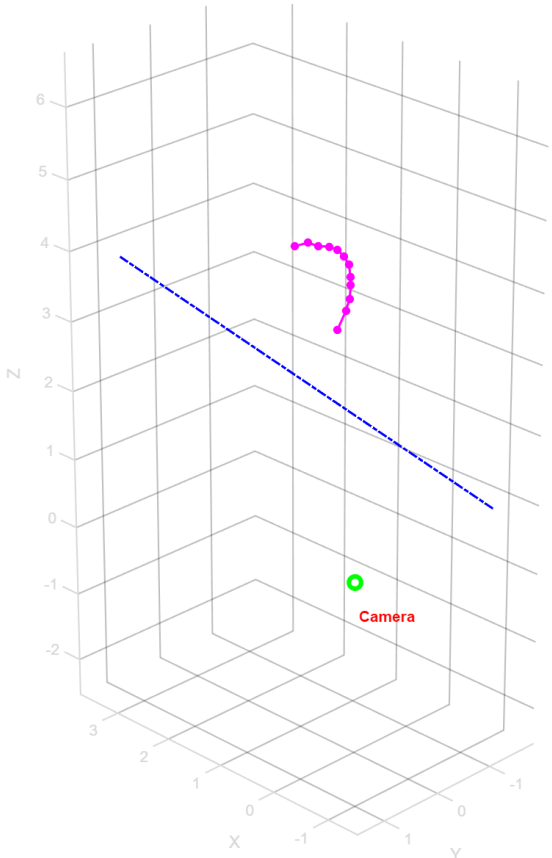
(a) Isometric View



(b) Bird's Eye View



(c) Worm's Eye View



(d) Rear-Side View